

Requirements for the flight computers for IREC

The DTEG has no dedicated avionics bay section that I found. The requirements are spread across the recovery, GPS, arming and airframe sections, and the main ones are below. "DTEG" is the 2026 DTEG v1.1 and "Rules" is the 2026 Rules & Requirements document.

Flight computers and redundancy

- Recovery systems must be independent and redundant, with separate arming switches, sensors or flight computers, power supplies, energetics and initiators (DTEG 6.9.1, p. 18). A dual-pole switch does not satisfy this (DTEG 6.9.2, p. 18).
- At least one recovery electronics subsystem must use a COTS flight computer (DTEG 6.10.1, p. 19). A COTS flight computer must serve as the official altitude logger and fire either the primary or redundant energetic system (DTEG 6.10.2 and 6.10.3, p. 19).
- You must carry a COTS barometric altimeter with onboard data storage. It can be a standalone unit or part of the COTS flight computer (Rules 2.5, p. 13).
- Any telemetry must come from the same sensor source as the official onboard log (Rules 2.5, p. 13).
- Active flight control systems must stay neutral until 6,500 ft AGL on 10k flights, or until burnout or max Q (DTEG 7.4, p. 24).

Wiring

- Safety critical wiring needs these (DTEG 6.12, pp. 19-20):
 - Cable management to prevent tangling (6.12.2.1, p. 19).
 - Labels or color coding (6.12.2.2, p. 19).
 - Strain relief at all connections (6.12.2.3, p. 20).
 - Connections secure against disconnection (6.12.3.1, p. 20).
 - No twisted-wire flight-essential connections (6.12.3.4, p. 20).
- Recovery components must have ejection gas protection (DTEG 6.5.1, p. 18).

Arming

- Energetics must be made safe until the rocket is in the launch position (DTEG 6.14.4, p. 21).
- All energetic device arming features must be externally actuatable and controllable (DTEG 6.15.1, p. 21).
- The arming features must be operable without a head within 100 mm of the rocket, while wearing a face shield and other PPE (DTEG 6.15.3 and 6.15.4, p. 22).
- They must be located so an inadvertent release won't impact the person arming (DTEG 6.15.5, p. 22).
- Arming must be verifiable in 76 dBA noise plus 20 mph wind (DTEG 6.16, p. 22).

- The rail must not block access to the arming electronics (DTEG 8.6.10, p. 27).

Power

- No LiPo, no rectangular or plastic-cased Li-Ion, no wet cell lead acid, and no nuclear batteries (DTEG 6.17.1, p. 22).

GPS

- Each independently recovered body needs a COTS GPS tracker (DTEG 4.2.1, p. 7).
- The tracker must be switched and powered independently of any flight computer, unless it is an integrated GPS flight computer (DTEG 4.2.3 and 4.2.3.1, pp. 7-8).
- A tracker that allows a custom name must include the Team ID number (DTEG 4.3.8, p. 9).

Airframe

- Vent the vehicle to prevent internal pressure damage (DTEG 8.1.1, p. 26).

Inspection, documentation and scoring

- At post-flight inspection you must open the electronics bays and bring the tools and equipment to read the altimeter. Expect no power or internet, and you have 45 minutes (Rules 2.5, pp. 13-14).
- Your report must record the altimeters used, whether each is COTS, and whether they are redundant (Rules 2.6.2.9.1, pp. 19-20).
- It must also include a figure and text showing the dual redundancy of the recovery electronics (Rules 2.6.2.9.2, p. 20).
- Your arming checklist must give detailed step-by-step procedures (Rules 2.6.2.9.5, p. 21).
- Significant damage to the avionics bay counts as excessive damage and fails recovery (Rules 2.8.1.4, p. 30).